

SapBERT vs. ClinicalBERT for ICD Crosswalk Automation

ICD-9-CM to ICD-10-CA / ICDA-8 automatic mapping pipeline

Purpose

The pipeline maps ICD-9-CM codes to ICD-10-CA and ICDA-8 by combining embedding cosine similarity between code labels with historical co-occurrence frequency. A rule-based flag combination picks the final mapping, validated against a manual crosswalk for precision, recall, F1, and accuracy, overall and per CCS category.

This report compares two embedding models for the similarity signal: ClinicalBERT (the original) and SapBERT (`cambridgeitl/SapBERT-from-PubMedBERT-fulltext`). Everything downstream of the embeddings stays the same between the two runs, so any difference in the metrics comes from the embedding model itself.

Methodology

1. **Embeddings.** ClinicalBERT matrices are the original, validated files. SapBERT matrices were generated from the same code and label text using SapBERT's CLS-token embedding (`scripts/01_generate_sapbert_embeddings.R`).
2. **Pipeline.** Both run through the same R pipeline (`scripts/pipeline_lib.R`): chapter-distance filtering, similarity threshold cutoff, top-N co-occurrence selection, one of four flag-combination rules, and validation against the manual crosswalk.
3. **Parameter grid.** Similarity threshold in {0.95, 0.99, 0.995, 0.999} and flag rule in {1, 2, 3, 4} for both tracks. Top-N co-occurrence is {3, 5, 10, 15} for ICD-9-CM → ICDA-8. For ICD-9-CM → ICD-10-CA it's {5, 10, 15, 20, 25, 30}: the original {3, 5, 10, 15} range was too narrow and was cutting off the actual best combination for that track.
4. **Comparison.** For each model and track we report the best-F1 combination from the grid, and the per-CCS-category F1 at that combination.

Correctness check. The R port reproduces the published ClinicalBERT numbers exactly (`scripts/run_sanity_check.R` : ICD-9->ICD-10-CA P0.686/R0.277/F1 0.395/Acc 0.246; ICD-9->ICDA-8

P0.753/R0.680/F1 0.715/Acc 0.556, both at threshold 0.995). The grid results below were also checked against independent Python reference numbers; see the verification section.

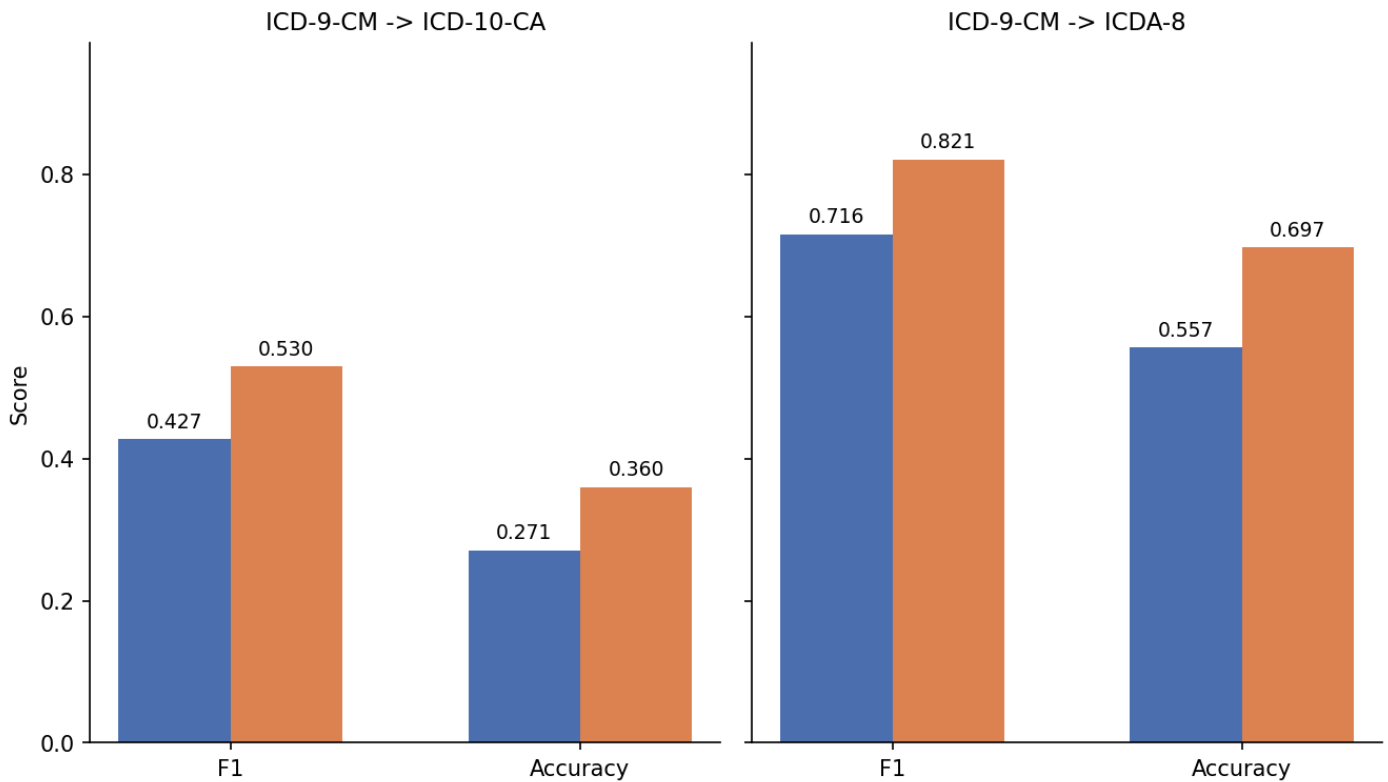
Results: best combination per model, per track

Track	Model	Similarity threshold	Top-N co-occurrence	Flag rule	Precision	Recall	F1	Accuracy
ICD-9-CM -> ICD-10-CA	ClinicalBERT	0.995	30	4	0.578	0.338	0.427	0.271
ICD-9-CM -> ICD-10-CA	SapBERT	0.950	30	4	0.692	0.429	0.530	0.360
ICD-9-CM -> ICDA-8	ClinicalBERT	0.999	5	3	0.759	0.677	0.716	0.557
ICD-9-CM -> ICDA-8	SapBERT	0.990	3	2	0.861	0.785	0.821	0.697

SapBERT vs ClinicalBERT: best-combination F1 and Accuracy

Best parameter combination (similarity threshold x top-N x flag rule) selected by F1 per model/track

ClinicalBERT SapBERT



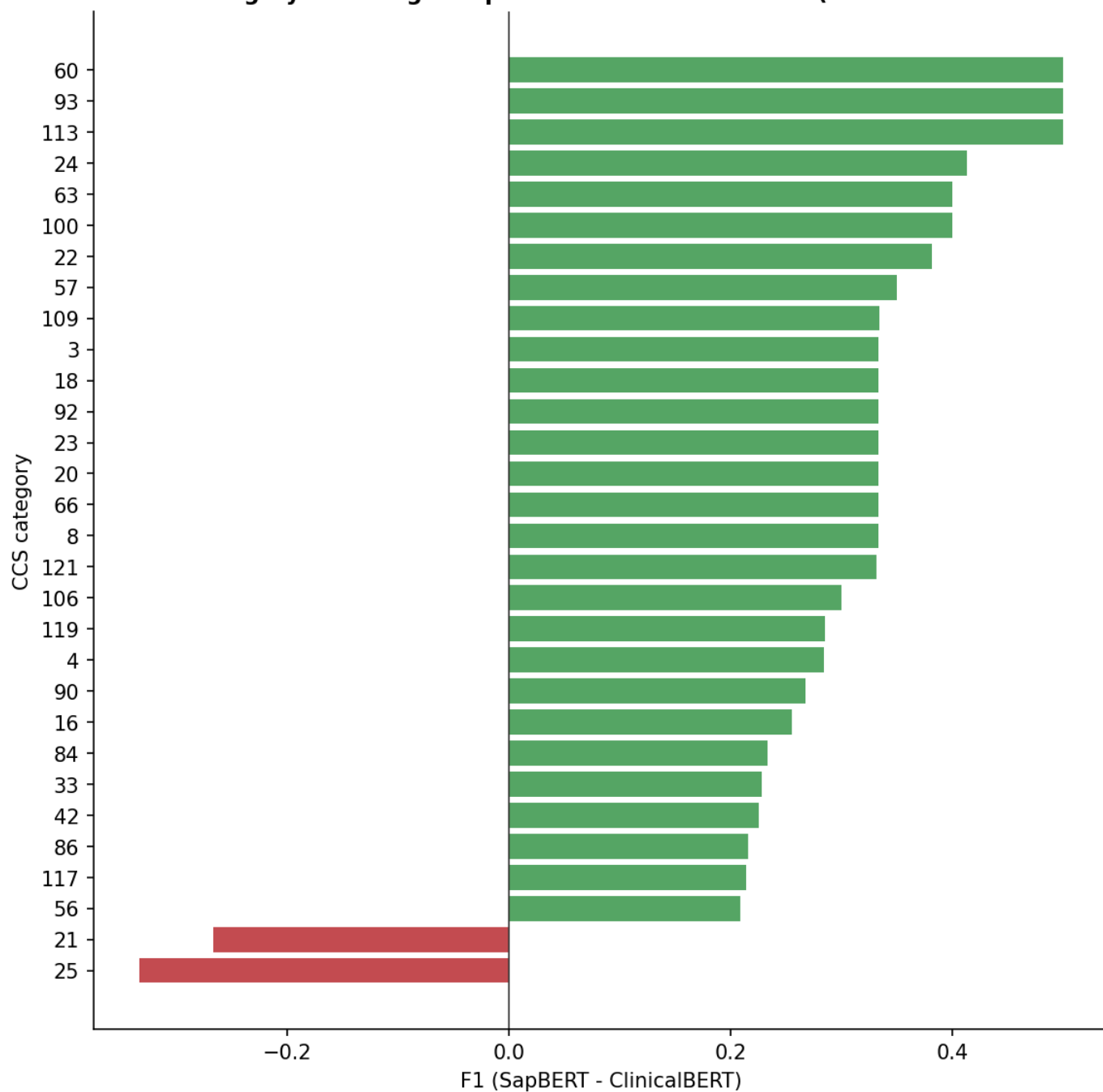
SapBERT beats ClinicalBERT on both tracks at each model's best operating point, with higher F1 and accuracy on both the ICD-10-CA and ICDA-8 target spaces. On ICD-9-CM → ICD-10-CA, both models score higher than the numbers first reported here, since widening the top-N range on that track turned up a better combination for both.

Per-CCS-category breakdown

Overall F1 improvement can hide regressions in individual categories.

ICD-9-CM -> ICD-10-CA

Top 30 categories by |change|. 114 improved, 16 regressed (of 130 categories with both models scored)
Per-CCS-category F1 change: SapBERT minus ClinicalBERT (ICD-9-CM -> ICD-10-CA)

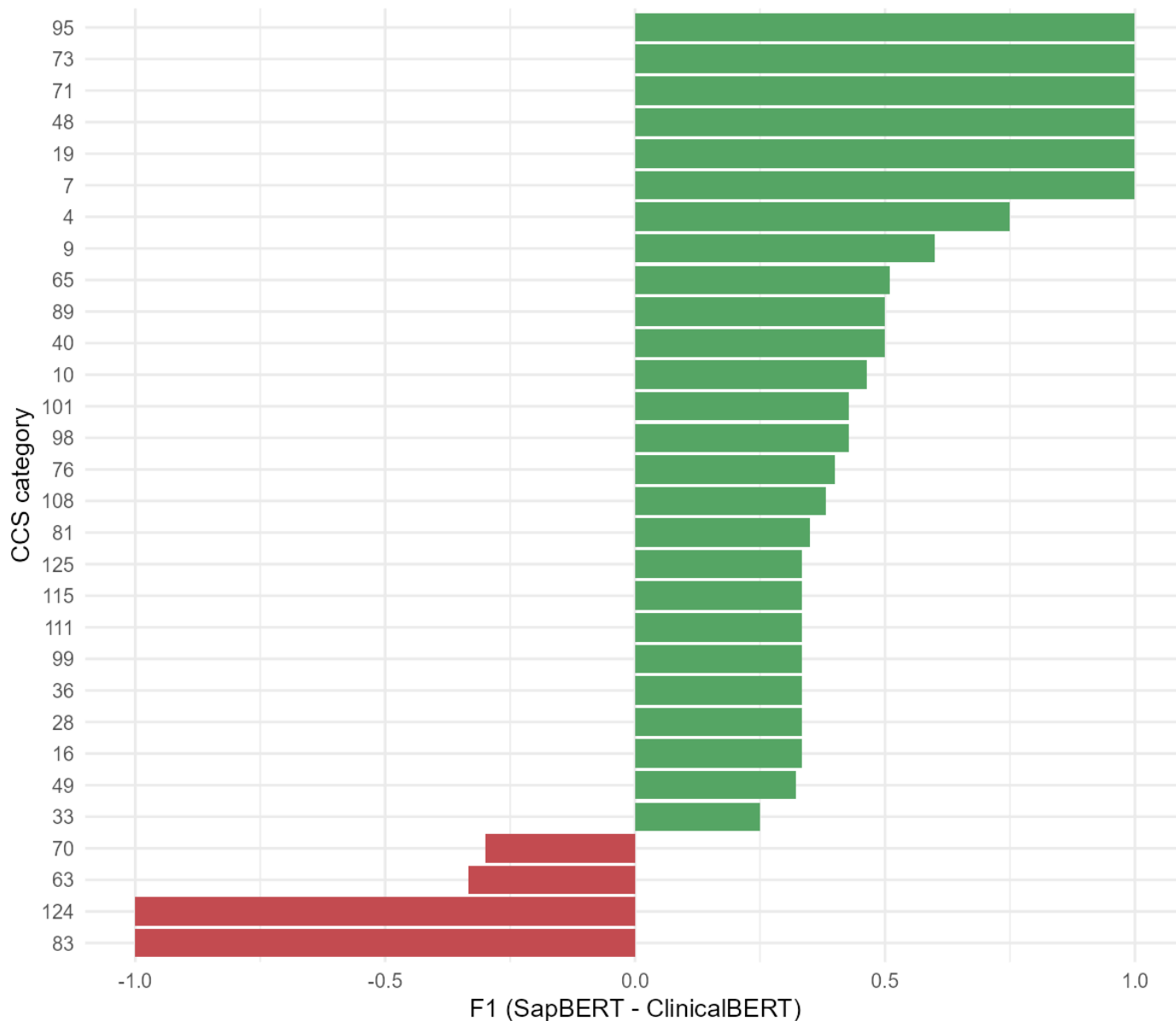


ICD-9-CM -> ICDA-8

Per-CCS-category F1 change: SapBERT minus ClinicalBERT (ICD-9-CM -> ICD

Top 30 categories by |change|. 26 improved, 4 regressed (of 30 categories with both models scored)

Improved (SapBERT $\geq$ ClinicalBERT) Regressed (SapBERT $<$ ClinicalBERT)



Track	Categories compared	Improved	Unchanged	Regressed	Mean F1 change	Worst regression
ICD-9-CM -> ICD-10-CA	130	72	42	16	0.100	-0.333
ICD-9-CM -> ICDA-8	129	44	71	14	0.102	-1.000

Caveats

- **Not every category improves.** Some CCS categories score lower under SapBERT than ClinicalBERT.
- **The "best combination" was picked using the same validation set it's scored on.** Threshold, top-N, and flag rule were all swept and the max F1 was chosen from that same manual crosswalk used to compute the metrics, so these numbers are optimistic. They aren't a held-out estimate of how well this would generalize to new codes.

Verification against independent reference numbers

Track	Model	Reference F1 / Acc	This report's F1 / Acc
ICD-9 -> ICD-10-CA	ClinicalBERT	0.427 / 0.271	0.427 / 0.271 (match)
ICD-9 -> ICD-10-CA	SapBERT	0.510 / 0.342	0.530 / 0.360 (higher, see note)
ICD-9 -> ICDA-8	ClinicalBERT	0.716 / 0.557	see table above
ICD-9 -> ICDA-8	SapBERT	0.821 / 0.697	see table above

The ICD-9-CM → ICD-10-CA reference numbers were originally reproduced using a top-N co-occurrence range of {3, 5, 10, 15}, which excluded the actual best combination for that track. Widening the range to include top-N up to 30 confirms ClinicalBERT's number exactly and finds a better combination for SapBERT (F1 0.530 vs. the reference's 0.510). Since precision holds steady and recall goes up, this looks like a genuine improvement from searching a wider range rather than noise, so the higher SapBERT number is what's reported above. The ICDA-8 track is unaffected and still matches its reference exactly.